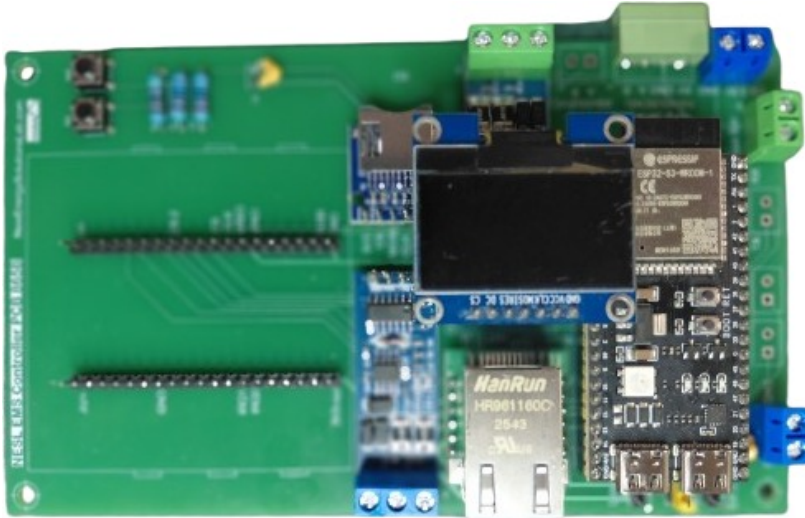


STREET-EMS

Front-of-Meter Feeder Control for the Village Minigrid Street Cabinet



Designing a village minigrid? **STREET-EMS** is the open-source feeder controller for the integration engineer building the **street pedestal cabinet** — coordinating breakers and contactors, metering each branch, and balancing distributed generation across the low-voltage feeder. Front-of-meter, peer-to-peer, and autonomous at the edge.

What Sets STREET-EMS Apart:

- ▶ **Cabinet-Ready Control** : drives contactors and breaker coordination from an onboard relay plus an 8-channel I²C SSR expansion bank for per-branch switching.
- ▶ **Open & Vendor-Agnostic** : built on the open-source meshEMS stack; integrates existing meters and DERs over Modbus, with no proprietary lock-in.
- ▶ **Edge-Native Balancing** : feeder metering and load balancing run locally on a dual-core ESP32-S3, with no dependence on cloud or central SCADA.

From a single street pedestal, STREET-EMS meters every branch, switches loads via contactors, and coordinates DERs — bringing intelligent, balanced power to the village minigrid feeder.

Intelligent

- ▶ Multi-branch metrology: V, I, P, Q, PF, import/export kWh
- ▶ Per-channel kW / kWh trip rules for load shedding
- ▶ Roadmap: SIP energy-session admission

Interoperable

- ▶ 2× RS-485 (Modbus) + CAN 2.0
- ▶ 8-ch I²C SSR for contactors / breakers
- ▶ WiFi / BLE & wired Ethernet

Resilient

- ▶ Full edge autonomy without backhaul
- ▶ Onboard + expansion relay control
- ▶ microSD logging | OLED status

Product Specifications

PRODUCT	
Product name	STREET-EMS (10Power X-EMS, supplied by NESL)
Model / PCB	NESL EMS Controller PCB 886B
Firmware	meshEMS / OpenAMI metering (open source, ~50k LOC middleware)
COMPUTE	
Processor	ESP32-S3-WROOM-1 dual-core Xtensa LX7 @ 240 MHz
Memory	16 MB Flash + 8 MB PSRAM
ASSET COMMUNICATION	
Serial buses	2× RS-485 Modbus: EVC + DTM/sensor ports 1× CAN 2.0
On-board metering	SPI poly-phase metering, up to 6 CT channels (CircuitSetup ATM90E32) V, I, P, Q, PF, freq
Local I/O	2× onboard SSR door-switch input 2 analog buttons
SUPPORTED MODULES & SENSORS (out of the box)	
Energy meters (Modbus)	DDS238 & CHINT CHD130 single-phase import/export kWh
Relay expansion	8-channel multiport I ² C SSR bank (PCF8574, <i>latest</i>) per-channel kW / kWh trip rules
Environment sensor	SHT20 (temperature / humidity, Modbus)
DER interoperability	SunSpec models 1 / 11 / 213 CAN (MCP2515)
NETWORKING	
LAN	10/100 Ethernet RJ45
Wireless	WiFi 802.11 b/g/n Bluetooth 5 LE
DISPLAY & STORAGE	
Display & storage	1.3 in OLED (SH1106) microSD card slot
POWER & ENCLOSURE	
Operating voltage	5 V DC – USB-C or screw terminals <i>[to confirm]</i>
Power / dimensions	<i>[to confirm]</i>
STANDARDS & POSITIONING	
Grid position	Front-of-meter (FTM) LV feeder street pedestal cabinet
Target standards	IEEE ISV (front-of-meter feeder) OpenAMI metrology
Capabilities	Peer-to-peer DER & feeder balancing contactor / breaker switching

Customizable with added hardware modules and open or licensed code stacks. Specifications are preliminary; items marked *[to confirm]* are to be confirmed on the production unit. STREET-EMS is an open-source X-EMS product supplied by New Energy Solutions Lab for 10Power.